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Unit “ 3 : Session “ 5 : SLO “ 2 | Evaluating a Postfix Expression

■ I. Algorithm to Evaluate a Postfix Expression

- Step 1: Scan the postfix expression from left to right.
- Step 2: If the symbol is an OPERAND, PUSH its value onto the stack.
- Step 3: If the symbol is an OPERATOR, POP two operands (first pop = operand2, second pop = operand1), compute operand1 OPERATOR operand2, and PUSH the result back.
- Step 4: Repeat until the expression ends.
- Step 5: The single value remaining on the stack is the result.
- Step 6: EXIT

■ II. Worked Example: 9 3 4 * 8 + 4 / -

- 9 -> push. Stack: [9]
- 3 -> push. Stack: [9, 3]
- 4 -> push. Stack: [9, 3, 4]
- * -> pop 4, 3; $3*4 = 12$; push. Stack: [9, 12]
- 8 -> push. Stack: [9, 12, 8]
- + -> pop 8, 12; $12+8 = 20$; push. Stack: [9, 20]
- 4 -> push. Stack: [9, 20, 4]
- / -> pop 4, 20; $20/4 = 5$; push. Stack: [9, 5]
- - -> pop 5, 9; $9-5 = 4$; push. Stack: [4]
- Result: 4